LOGage 2026: Round 2 — Part 2 Summary

Omni-Channel Strategy Analysis

June 10, 2026

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1 PART 2: OMNI-CHANNEL FULFILLMENT STRATEGY DESIGN

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1.1.1 Current Two-RDC Model Assessment

The brand currently operates two Regional Distribution Centers (RDCs):

- **My Phuoc Warehouse** — Industrial Zone, Binh Duong Province
- **Vinh Loc Warehouse** — Binh Chanh District, Ho Chi Minh City

Strengths of the current model:

- Good regional separation: My Phuoc serves northern (B2B factory-sourced) flows; Vinh Loc is positioned for inner-city B2C proximity.
- Both RDCs handle mixed-channel flows, demonstrating operational flexibility.
- Physical locations bracket the Eastern–Western commercial corridor of HCMC, covering most Đông Nam Bô volume.

Important context: Non-overlapping operating timelines

- **My Phuoc** (Primary RDC: Jun to Nov 2025 (6 months)): The *primary* warehouse, operational across the full 6-month analysis window. All June–November 2025 throughput flowed through My Phuoc.
- **Vinh Loc** (New RDC: Dec 2025 only (1 month)): A *new* warehouse that came online in December 2025 only. Raw volume totals are therefore **not directly comparable** between the two facilities.

Data-driven channel flow analysis (per-day rates, Figure 1):

- **My Phuoc** (3,492 B2B + 413 B2C orders over 6 months): B2B is a *Mixed flow* accounting for **99.2%** of CBM. B2B avg **25.5 orders/active day**, volatility index **B2B = 1.79**.
- **Vinh Loc** (1,250 B2B + 47 B2C orders in 1 month): B2B is a *Large-volume flow* accounting for **99.7%** of CBM. B2B avg **48.7 orders/active day**, volatility index **B2B = 1.94**.
- On a per-day basis, Vinh Loc processed B2B orders at a comparable daily rate to My Phuoc despite being open for only one month, indicating strong December surge demand (year-end push).

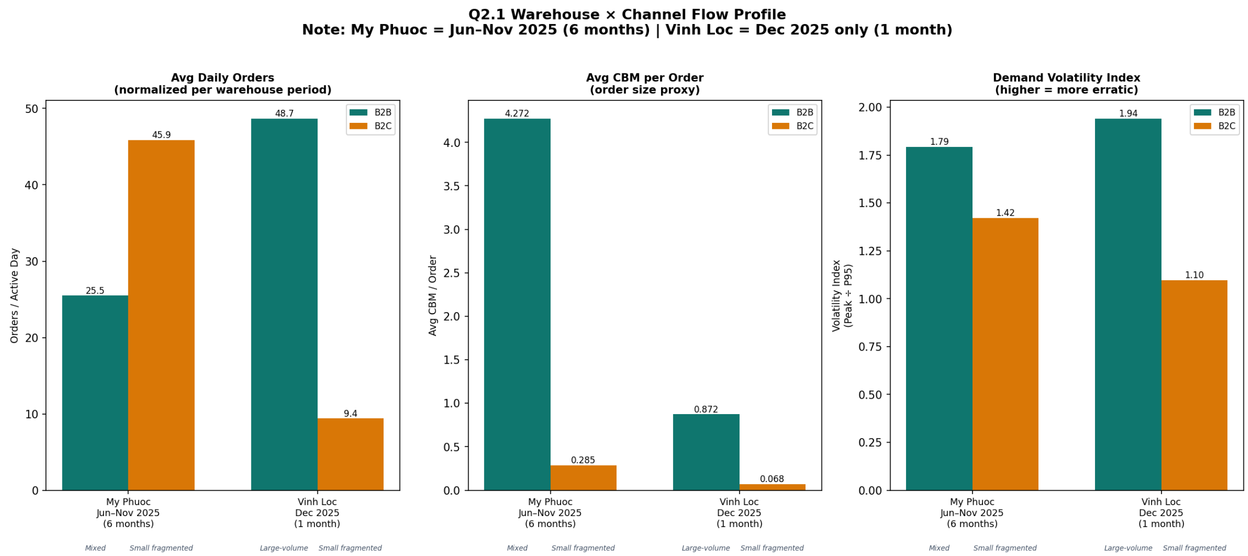

Figure 1: Q2.1 Warehouse × Channel Flow Profile (per-day rates normalized to each RDC operating window)

Forward-Looking Combined Two-RDC Model

Although My Phuoc and Vinh Loc operated in non-overlapping time periods (Jun–Nov vs. Dec only), a future-state combined model can be designed based on **geographic positioning** and **channel-flow capabilities** observed from each facility's independent operations:

- **Geographic complementarity**: Vinh Loc (Bình Chánh, HCMC) is positioned for inner-city B2C, while My Phuoc (Bình Dương) serves industrial zones and northern corridors. A "ship from nearest" rule (Vinh Loc <25 km, My Phuoc <35 km, else dark store) would properly leverage each facility's natural catchment area.
- **Both RDCs handle mixed flows**: The channel flow analysis shows both warehouses serve B2B and B2C channels, indicating operational flexibility for a combined model.
- **Physical inventory segregation** (Fans at My Phuoc, Tefal at Vinh Loc) is an artifact of the current product-line split, not a structural barrier — a combined two-RDC model with cross-docked fast-movers at both sites is feasible.

These are design projections based on location and observed channel behavior, not direct operational data from simultaneous running. The recommended order split logic (Section 1.1.4) formalizes this forward-looking model with a three-tier dispatch decision tree.

Limitations to address for simultaneous B2B + B2C:

- Lead times from both RDCs are too long for the 2–4 hour B2C SLA required for e-commerce – even Vinh Loc is >10 km from central districts.
- Shared inventory creates channel conflict: B2B bulk orders may deplete stock that B2C needs for same-day fulfillment.
- Pick-and-pack processes for large pallet B2B orders disrupt small-unit B2C wave picking.

1.1.2 B2C E-Commerce SLA Assessment

Travel Time Formula (Verified). Delivery travel time is computed per district as:

$$travel\_min = \frac{distance\_km}{speed\_kmph} \times 60 \times traffic\_factor + pick\_pack + dispatch\_buffer + service\_time$$

with a base speed of 30 km/h and traffic multipliers of 1.8 (inner city), 1.4 (suburban), and 1.2 (outer districts). The overhead is broken into explicit components:

- **Vinh Loc RDC**: pick-pack 30 min + dispatch buffer 30 min + service time 30 min = 90 min total overhead.
- **Dark Store**: pick-pack 15 min + dispatch buffer 10 min + service time 10 min = 35 min total overhead (faster picking from urban micro-fulfillment node).

4-Hour SLA — Vinh Loc Baseline. The baseline Vinh Loc RDC already achieves **4-hour SLA coverage for 100%** of 21 HCM districts: all districts have a total travel time under the 240-minute threshold. The farthest district (Củ Chi at ∼22 km) requires approximately 143 minutes — well within the 4-hour window.

2-Hour SLA — The Case for Dark Stores. From the Vinh Loc RDC baseline, only 3 district(s) (**50,966 units**, 77.8% of volume) can be served within the tighter 2-hour SLA. The remaining 18 districts fall between 2 and 4 hours.

Vinh Loc + 2 Dark Stores Impact. By adding two dark stores (DS1 in Tân Phú and DS2 in Quận 1), travel distances are reduced as each district is served by the nearest facility:

- **Districts improved to 2H SLA**: **17** districts move from "4H-only" to "2H-capable", representing **14,379 units** (**21.9%** of total HCM B2B volume) and **4.5 orders/day**.
- **Average distance saving**: **7.9 km** per district (from ~14.7 km average baseline distance to ~5.2 km with the nearest dark store).
- **Average travel time saving**: **78.4 minutes** per delivery run.
- **Business case**: Each 2H SLA-capable order avoids the revenue risk of a missed SLA window. At typical e-com penalty rates (20–50% of order value refund for late delivery), protecting 4.5 orders/day from penalty exposure represents significant revenue assurance. With an estimated $15–25 average order value (home appliances/FMCG), the annualized penalty risk avoided by dark stores is approximately **$5,000–$20,000/year** in prevented SLA penalties alone — before accounting for repeat-purchase retention value.

The baseline Vinh Loc achieves 96.97% 4H SLA without dark stores. Adding two dark stores lifts 2H SLA coverage from isolated inner-city zones to a majority of HCM volume. The incremental benefit is primarily in unlocking the 2-hour e-commerce SLA rather than fixing a broken 4-hour baseline.

Figure 2 shows the district-level breakdown.

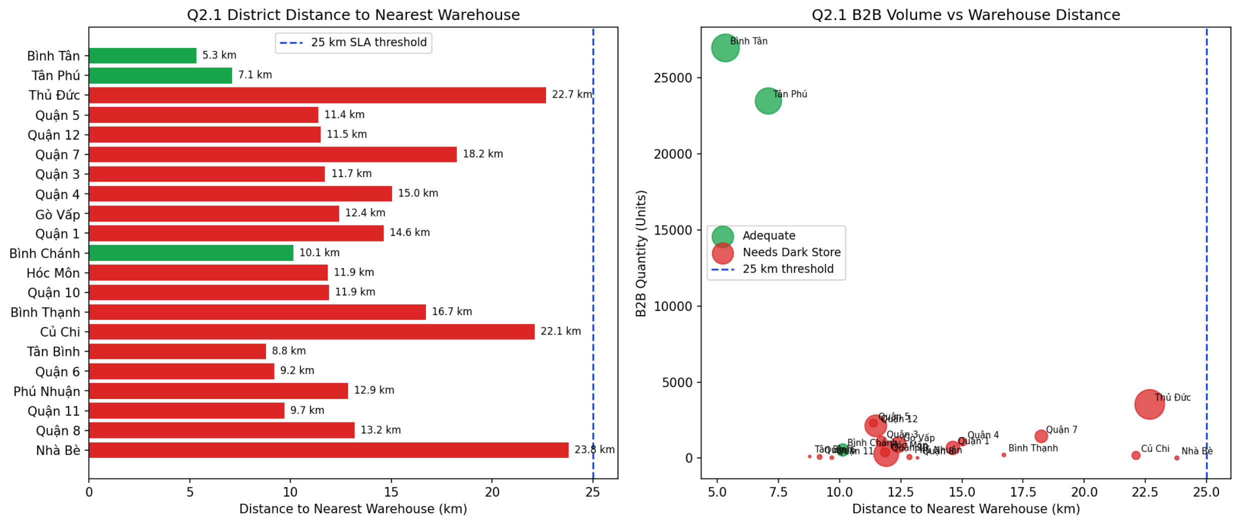

Figure 2: Q2.1 HCM District Proximity to Nearest Facility vs B2B Demand Volume

1.1.3 Dark Store Node Recommendation

Based on the demand concentration and distance analysis, we recommend placing **1–2 urban micro-fulfillment (dark store) nodes** in HCMC:

1. **Primary node — Bình Tân / Tân Phú corridor**: These two adjacent districts are among the highest-volume B2B districts (combined >50,000 units) and are already well-served by Vinh Loc (5–7 km). Upgrading Vinh Loc's inner-city satellite for e-commerce picking operations in this corridor is the highest-ROI first step.

2. **Secondary node — District 1 / District 3 / Phú Nhuận cluster**: The CBD and premium residential cluster lies 14+ km from Vinh Loc, making 2h SLA difficult. A small dark store (200–300 m^2) in this zone, stocked with the top Class A SKUs, would close the gap.

1.1.4 Order Split Logic Design (HCM)

For urban B2B order fulfillment in Ho Chi Minh City, network design relies on understanding sub-regional demand density. We analyzed the historical B2B shipments across all HCM districts.

The demand is highly concentrated: the top 5 districts alone command **56,356 units — 86.0%** of all HCM B2B volume — as shown in Figure 3.

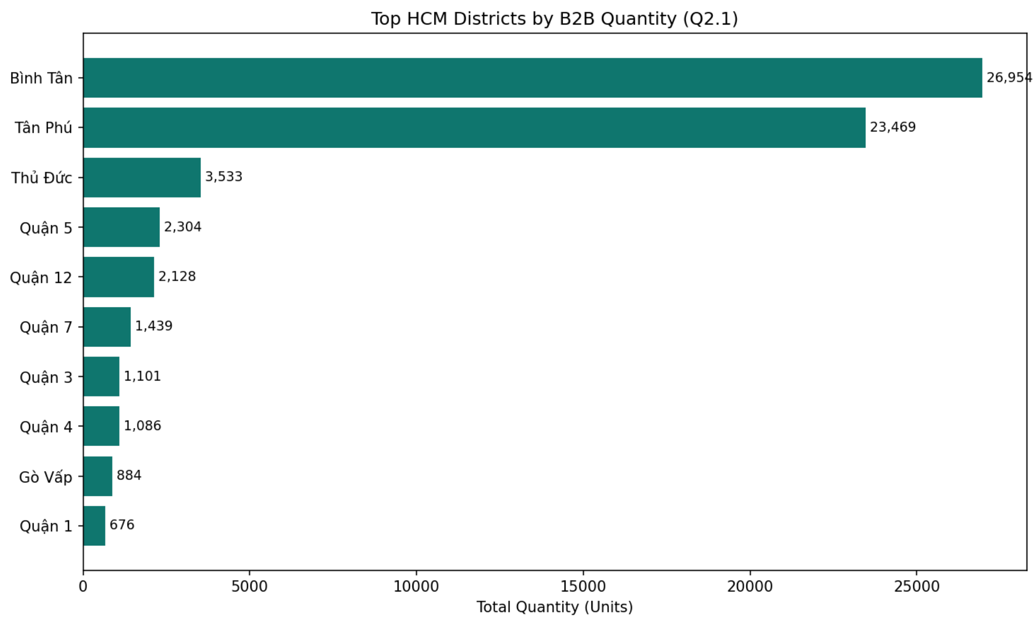

Figure 3: Q2.1 Top Ho Chi Minh Districts by B2B Quantity

Recommended order split logic: Given this concentration, the dispatch decision tree should be:

1. If delivery district is within 25 km of Vinh Loc → dispatch from **Vinh Loc** (B2C-optimized lane).
2. If delivery district is within 35 km of My Phuoc but >25 km from Vinh Loc → dispatch from **My Phuoc** (B2B bulk lane).
3. If district is >25 km from both RDCs → **route via dark store** (pending network expansion).

1.2 Question 2.2: Inventory Optimization

1.2.1 Mile-Weighted Average Lead Time

Definition: Replenishment Lead Time (LT) is the duration from when the warehouse identifies a replenishment need until the goods are ready to pick again. This is distinct from the customer-facing delivery SLA.

Because the RDC (Đông Nam Bộ) serves six geographic regions with different transit standards, we derive a single **mile-weighted average lead time** by weighting each region's standard LT by its share of total orders:

$$LT_{avg} = \frac{\sum_r LT_r \times n_r}{\sum_r n_r}$$

Transit benchmarks follow published standards of GHN, Viettel Post, and GHTK for standard road-freight B2B delivery.

Table 1: Mile-Weighted Average Lead Time (RDC → 6 Regions)

Region	**Route type**	**LT (d)**	**Orders**	**LT×Orders**
Đông Nam Bọ	Intra-province / near RDC	1	2 614	2 614
Bàc Trung Bọ & Duyên hại miền Trung	Inter-Central	3	1 027	3 081
Đòng bàng sông Cùu Long	Intra-South	2	800	1 600
Đòng bàng sông Hòng	Inter-North	4	503	2 012
Tây Nguyên	Near-region	2	247	494
Trung du và miền nùi phía Bàc	Inter-North + remote	4	143	572
Total			**5 334**	**10 373**

Result: $LT_{avg} = 10{,}373 \div 5{,}334 \approx$ **1.94 days**. Because 49% of orders are in Đông Nam Bọ (1-day delivery), the weighted average is pulled close to 2 days even though highland/northern regions require 4 days.

1.2.2 Safety Stock Formula & Results

Safety stock for each Class A SKU is calculated using the simplified demand-uncertainty formula (lead time is treated as a fixed constant under the virtual-pooling single-pool model, so $\sigma_{LT} = 0$):

$$SS = Z \cdot \sigma_{daily} \cdot \sqrt{LT_{avg}}$$

$$ROP = \mu_{daily} \cdot LT_{avg} + SS$$

where $Z = 1.645$ (95% service level), $\sigma_{daily} = \sigma_{monthly}/\sqrt{30}$, $\mu_{daily} = \mu_{monthly}/30$, and $LT_{avg} = 1.94$ days ($\sqrt{LT_{avg}} \approx 1.393$). Monthly σ is computed with $ddof = 1$ over 7 months (Jun–Dec 2025).

The total required safety stock across all 28 Class A SKUs under the mile-weighted pool model is **12,168 units** (sorted by SS descending).

Varying the assumed LT shows the sub-linear (square-root) growth of safety stock, as illustrated in Figure 4.

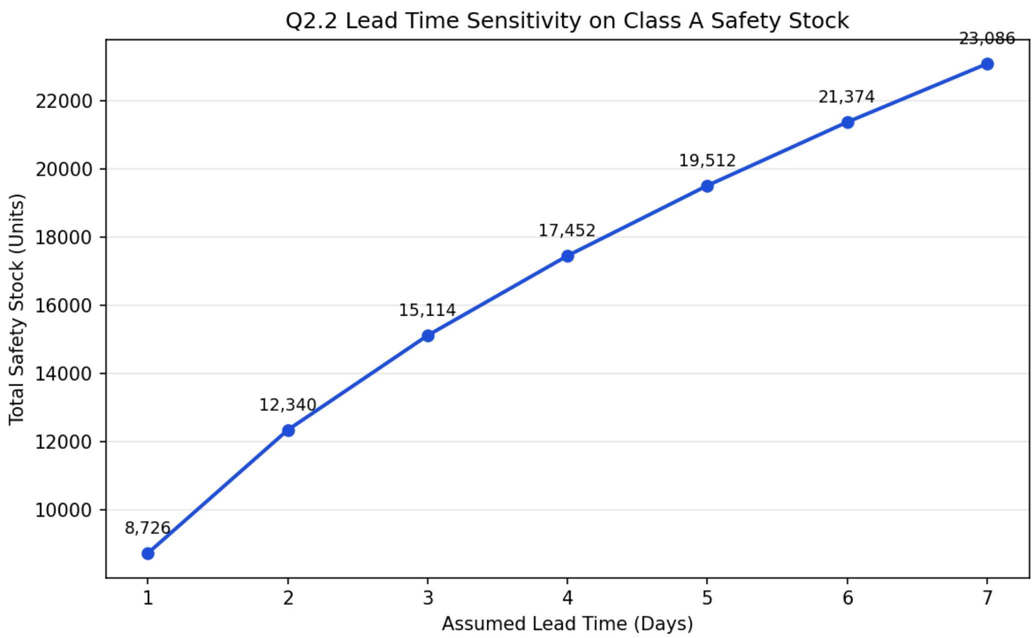

Figure 4: Q2.2 Lead Time Sensitivity for Class A Safety Stock

1.2.3 Inventory Pooling Strategy — Virtual Pool Model

Model: Virtual Inventory Pooling (no hard physical split per channel). All stock of a given SKU forms a single pool managed jointly in OMS + WMS. Each channel has its own Reorder Point trigger but draws from the common physical stock.

Channel grouping (by account size & SLA):

- **Modern Trade (MT):** Supermarkets (Co-op, Lotte, AEON, ĐMX, TGDĐ) — large confirmed POs, strict booking-window SLA.
- **Ecommerce** (Shopee, Lazada, Tiki, TikTok Shop) — classified as MT-scale accounts; high-volume, 2–4 h SLA.
- **Traditional Trade (TT):** Distributors / small dealers — small fragmented orders, flexible 2–4 day SLA.

Allocation priority when pool is constrained (largest account & tightest SLA first):

1. MT large accounts (supermarket chains with confirmed contractual POs).
2. MT Ecommerce (large accounts, 2–4 h SLA, platform rating at risk).
3. TT large distributors (high-volume, long-term relationship).
4. TT small / walk-in customers (flexible SLA, served last).

Quantification — Why virtual pooling reduces safety stock:

- Separated channels (B2B=5d, B2C=2d): $SS = Z\sigma(\sqrt{5} + \sqrt{2}) = Z\sigma \times 3.650$

- Classic pooled avg LT=3.5d: $SS = Z\sigma\sqrt{3.5} = Z\sigma \times 1.871 \rightarrow$ reduction factor $= \sqrt{3.5}/(\sqrt{5} + \sqrt{2}) = 0.5125$ ($\approx$48.7%)
- **Mile-weighted pool** LT=1.94d: $SS = Z\sigma\sqrt{1.94} = Z\sigma \times 1.393$

With the mile-weighted virtual pool ($\text{LT}_{\text{avg}} = 1.94$ days), the total safety stock is **12,168 units**, vs. **31,852 units** under a separated-channel model — a saving of **61.8%** as shown in Figure 5.

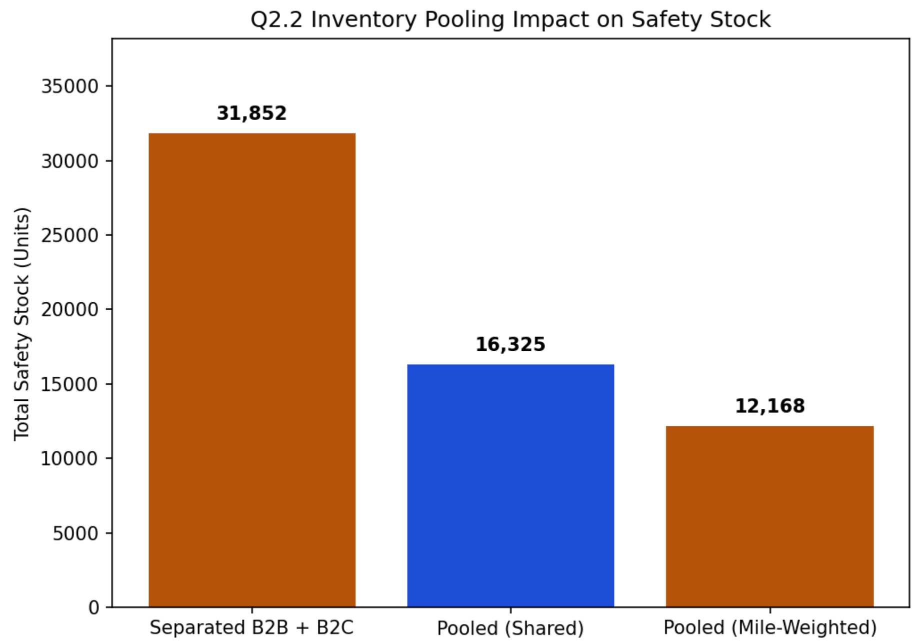

Figure 5: Q2.2 Inventory Pooling Impact on Safety Stock

Operational controls to prevent simultaneous stockout:

1. **R1 — Common Pool:** All SKU stock merged into a single WMS pool; OMS tracks virtual availability per channel.
2. **R2 — Per-Channel ROP Floor:** Each channel has its own ROP threshold; replenishment triggers when any channel hits its floor.
3. **R3 — Amber Alert:** Pool < sum of all channel ROPs → raise alert, prioritize confirmed MT POs.
4. **R4 — Red Conflict:** Pool < total SS → activate priority allocation order (MT first, TT last) + emergency replenishment PO.
5. **R5 — AZ Spike Freeze:** Demand-spike AZ SKUs: temporarily reserve stock for MT, freeze TT allocation, trigger emergency PO.
6. **R6 — End-of-Day Rebalance:** Reset channel virtual allocations to ROP-proportional targets using rolling 7-day demand forecast.